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Application of 3.5-Inch Small-Diameter Well Coiled Tubing Workover Technology in South Sulige

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Abstract

In the middle and late stages of oilfield development, during oil and gas field stimulation operations, and due to geological factors, oil and gas well casings may be damaged. Casing damage restricts the normal production of oil and gas wells and may even lead to serious consequences such as wellbore overflow. Timely and effective plugging and reinforcement of damaged casings are crucial technical measures to ensure the normal production of oil and gas wells. This paper conducts research on the 3.5in casing cable-transmitted propellant-powered patching tool. It introduces the principle and technical characteristics of the propellant-powered patching tool; optimizes the design of the expansion cone, anchoring structure, and tool parameters through finite element simulation; finally, completes the ground test of the tool and the field application of the technology and process. After patching, the full wellbore pressure test is conducted at 50MPa, with pressure stabilized for 10min and a pressure drop of 0.7MPa. Field applications show that the tool is reasonably designed, and it is the first successful application of cable transmission for such tools.

Introduction

The 3.5-inch slim-hole high-pressure casing patch technology features excellent sealing, high pressure-bearing capacity, and good plasticity. It has been applied successfully and maturely in the Su-nan Block of PetroChinaChangqing Oilfield. To date, during workover operations, the technology has repaired the casings of gas wells while meeting the requirements for secondary fracturing. Two wells were successfully tested to 55 MPa, achieving a 100 % patch success rate. The mature application of this technology has broad implications for the promotion of slim-hole casing repair processes. As oil and gas field development enters the middle and late stages, casing damage impairs wellbore integrity and has become one of the key factors restricting oil and gas field development, particularly in mature oilfields such as Daqing Oilfield, Liaohe Oilfield, Shengli Oilfield, and Zhongyuan Oilfield [1–5]. In recent years, casing damage has also occurred frequently in new oil and gas fields: for example, horizontal well casing collapse in Southwest Oil & Gas Field, casing leakage due to corrosion in Changqing Oilfield gas wells, and casing deformation caused by poor cementing quality in Tarim Oilfield [6–8]. In addition, engineering factors can also lead to casing damage, such as wellbore water/steam injection, acid fracturing, and mis-perforation [9–11].

Currently, casing damage repair methods mainly include casing reshaping, casing sealing and reinforcement, casing retrieval, and chemical plugging [12–13]. Casing sealing and reinforcement can be further divided into hydraulic sealing reinforcement and propellant-powered patching. This paper focuses on the propellant-powered patching technology and its field application.

Propellant-Powered Casing Patching Technology

Tool Structure

The 3.5in casing propellant-powered patching system consists of three subsystems: the power system, mechanical conversion system, and patching tube system. These subsystems are closely connected and interact with each other. The design of the patching tube's sealing and anchoring device must match the parameters of other subsystems to meet the design requirements.

In Changqing Sulige Gas Field, 3.5in casing wells require stimulation fracturing during production. Unlike conventional fracturing methods, fracturing in Sulige Gas Field is conducted directly inside 3.5in casings, which requires the sealed pressure-bearing capacity of the patched casing to reach 60MPa (the fracturing pressure). To meet this requirement, a composite sealing and anchoring structure with alternating expandable rubber materials and multi-ring frameworks was designed (see Fig. 1). The surface of the anchoring structure is coated with expandable rubber of a special structure. Supported by the multi-ring frameworks, the sealing effect of the expandable rubber is significantly improved; meanwhile, the multi-ring frameworks can better support the expandable rubber when expanded. Additionally, the multi-ring frameworks require a small expansion force, providing favorable conditions for the design of the 3.5in small-sized propellant-powered patching tool.

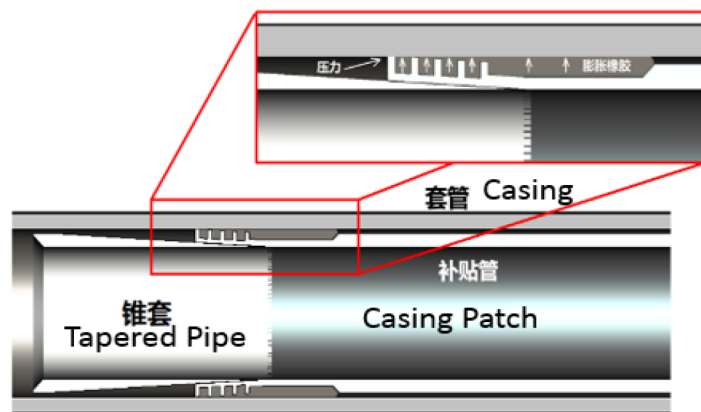


Figure 1—Schematic Diagram of Patching Tube Anchoring Structure

Tool Principle

The principle of the cable-transmitted propellant-powered patching technology is as follows: The cable supplies power to the igniter, which ignites the composite propellant carried by the system. The high-pressure gas generated by the combustion of the composite propellant drives the relative movement between the piston and the piston cylinder to do work, converting chemical energy into mechanical energy. This causes the taper sleeves at both ends of the patching tube to move relatively and insert into the patching tube, radially expanding the anchoring structures at both ends of the patching tube. As a result, the metal at both ends of the patching tube expands and anchors to the inner wall of the casing, forming an interference fit to achieve anchoring and sealing (the first seal).

The outer surface of the downhole patching tube is coated with expandable rubber, which reacts with well fluid and gradually expands to seal the annular gap between the casing and the patching tube, forming

the second seal. The patching tube isolates the casing-damaged area or the perforated section that needs plugging. The propellant-powered patching tool and patching tube system are shown in Fig. 2.



Figure 2—Assembly Diagram of Propellant-Powered Casing Patching Tool

Tool Parameters

According to the outer diameter requirements of downhole cable-transmitted tools for 3.5in casings, a propellant-powered patching tool for patching tube installation was designed. Its design indicators are shown in Table 1.

Table 1—Basic Parameters of 3.5in Casing Propellant-Powered Patching System

System Category	Technical Parameter			
	Casing Outer Diameter (in/mm)		Casing Inner Diameter (in/mm)	
Basic Parameters of Downhole Casing	3.5/88.9		2.922/74.22	
Parameters of Propellant-Powered Installation Tool	Outer Diameter of Installation Tool (in/mm)	Length of Installation Tool (in/mm)	Output Pull Force (lb/kN)	Stroke of Output Pull Force (in/mm)
	2.598/66	118.11/3000	60742/270	10.5/267
Parameters of Power Source	Temperature Resistance of Composite Propellant: 150°C/24H			

Technical Characteristics and Application Scope

The cable-transmitted propellant-powered patching technology has the following three characteristics:

High Efficiency: The cable-transmitted patching tool ensures high transmission efficiency. Compared with coiled tubing or tubing (drill pipe) transmission, it not only offers faster transmission speed but also eliminates the need for surface equipment such as coiled tubing trucks and workover rigs. Meanwhile, the technology has a built-in power source: propellant combustion is activated by electric ignition. Compared with drop bar (pressure) ignition and hydraulic expansion, this process is more reliable and does not require surface equipment such as pump trucks. Among all patching technologies, this technology uses propellant combustion to drive the tool, resulting in the shortest construction time.

High Pressure Resistance: The technology adopts a dual-seal anchoring structure with excellent pressure-bearing performance. The maximum pressure-bearing capacity in ground tests can reach 58.5MPa (8500psi).

Retrievable: By milling the anchoring structure at the upper end of the patching tube, lowering an external taper to fish the patching tube, and pulling out the patching tube, the wellbore diameter can be restored without affecting subsequent development measures.

Due to the use of cable transmission, this technology is applicable to vertical wells or deviated wells with a deviation angle of less than 45°.

Surface Pressure Test

A 1-m-long patch tube was used for a surface patch test to evaluate overall system safety and reliability. The patched sample was then pressure-tested to verify sealing performance. The test setup is shown in Figure 1, and equipment parameters are listed in Table 2.

Table 2—Equipment Parameters

NO.	Name	let	value
1	Casing	OD (in/mm)	3.5/89
2		ID (in/mm)	2.922/74.21
3	Patch tube	OD (in/mm)	2.677/68
4		ID (in/mm)	2.2/56
5	Anchor device	OD (in/mm)	2.677/68
6		ID (in/mm)	2.13/54
7	RB89 installation tool	OD (in/mm)	2.598/66
8		length (in/m)	116.1/2.95
9		Ignition current (A)	0.8 ~ 1.3

Simulation Calculation

Finite element simulation was used to analyze the internal and external pressure-bearing performance of the sealing and anchoring structure. The calculation results show that under internal pressure, the anchoring structure deforms and expands radially outward; supported by the outer casing, it meets the internal pressure-bearing requirements. However, under external pressure, the anchoring structure deforms and contracts radially inward, failing to meet the internal drift diameter requirement. To meet the downhole engineering design requirement of withstanding 50MPa external pressure, the taper sleeve guide section was added, which solves the external pressure-bearing problem of the anchoring structure (see the finite element calculation results in Fig. 3–2).



Figure 3—Operating Procedures

Based on the above analysis, the pressure-bearing strength design requirements of the patching tube's sealing and anchoring structure were verified through test experiments, as shown in Table 2.

Ground Tests

Ground tests were conducted based on the simulation results, including press tests, ground linkage tests, and pressure-bearing tests.

Ground Press Test and Pressure-Bearing Test

Ground press test tooling was fabricated, and its schematic diagram and physical diagram are shown in Fig. 3. After machining, the patching tube's anchoring mechanism was coated with expandable rubber. The expansion cone was inserted into the anchoring mechanism of the patching tube; a pressure block was placed

on top to increase the press stroke, and a spacer block was placed at the bottom to center the patching tube and facilitate subsequent welding.

During the test, the press applied an axial force of 250kN to the pressure block. The pressured block pushed the expansion cone downward, which caused the patching tube to expand radially until it attached to the inner wall of the casing, simulating the completion of patching on the ground.

End plates and pressure test joints were welded to both ends of the sample to form a pressure test specimen for internal pressure testing (see Fig. 4). The test results showed that the 3.5in casing patching simulation specimen could withstand an internal pressure of 55MPa with no leakage observed within 15 minutes.



Figure 4—Schematic Diagram of Test Tooling (Left), Physical Diagram (Middle) and Test Process (Right)

Ground Linkage Test and Pressure-Bearing Test

Based on the results of the ground press simulation patching and pressure-bearing tests, a ground linkage test of the propellant-powered patching tool was conducted. The propellant-powered patching tube installation tool was used to simulate downhole conditions and complete the installation of a 1m-long patching tube on the ground, to verify the safety and reliability of the entire system. The physical diagram of the propellant-powered patching tool and casing is shown in Fig. 5.



Figure 5—Pressure Test Specimen

The assembled test device was placed at the designated location in the test range, and a protective casing was added outside the test device to ensure test safety (see Fig. 6). After ignition, the system was left to stand for 10min to allow the propellant-powered patching tool to complete the patching tube installation. Inspection confirmed that the propellant-powered patching tool worked normally, and the patching tube was successfully installed on the inner wall of the 3.5in casing (see Fig. 7).



Figure 6—Assembly Diagram of Propellant-Powered Casing Patching Tool and Casing



Figure 7—Test Device

End plates and joints were welded to both ends of the patched sample to form pressure test specimens for internal and external pressure testing (see Fig. 8 and Fig. 10). The test results showed that the 3.5in casing patching simulation specimen could withstand an internal pressure of 61MPa (no leakage within 15 minutes) and an external pressure of 38MPa (no leakage within 15 minutes). The pressure-bearing curves are shown in Fig. 9 and Fig. 11 (the abscissa represents time, and the ordinate represents pressure).



Figure 8—Inner Wall of Casing After Test



Figure 9—Internal Pressure Test

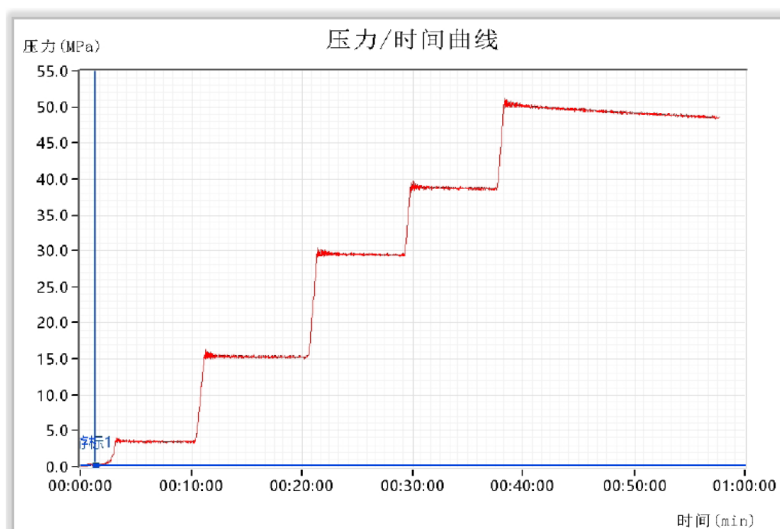


Figure 10—Internal Pressure Test Curve



Figure 11—External Pressure Test

Field Tests

Construction Procedures

1. Coiled tubing wall cleaning: Run the coiled tubing (with a rotary nozzle) to the to-be-patched position, start the pump to flush the pipe wall for at least 20m above and below the to-be-patched depth, ensuring the pipe wall in the to-be-patched section is clean and scale-free.
2. Simulated wellbore drifting: Assemble the drifting tool string: Cable Head + Magnetic Positioning (CCL) + Patching Power Tool + Customized Drifting Gauge + Guide. Run the tool string, and monitor the CCL signal and tension changes at all times to prevent sticking or jamming. The running speed shall not exceed 2000m/h, and the pulling speed shall not exceed 4000m/h.
3. Patching operation: Assemble the installation patching tool string: Cable Head + Magnetic Positioning (CCL) + Patching Power Tool + Customized Patching Tube + Guide. Test the circuit and signals to ensure they are normal. Run the tool string at a speed not exceeding 2000m/h to 10m below the patching position. Position the tool by pulling it up, energize for ignition, and let it stand for 10min. Then, lower the tool to check if setting is completed. After setting, pull out the tool string to finish patching.
4. Pressure test: Connect the surface pipeline, and the pump truck applies pressure stepwise to the specified pressure, stabilizing the pressure for 15min. The pressure test is qualified if no anomalies are found.

Application Cases

After laboratory verification, this technology was tested in the field in the Sunan Gas Field, achieving good application results.

The first test well was Well XXX-01. Logging analysis comprehensively determined that there was a leakage at 1794.55m. The leakage point at 1794.55m and the surrounding casing surface damage area were isolated by patching to seal the leakage point (see Fig. 12).

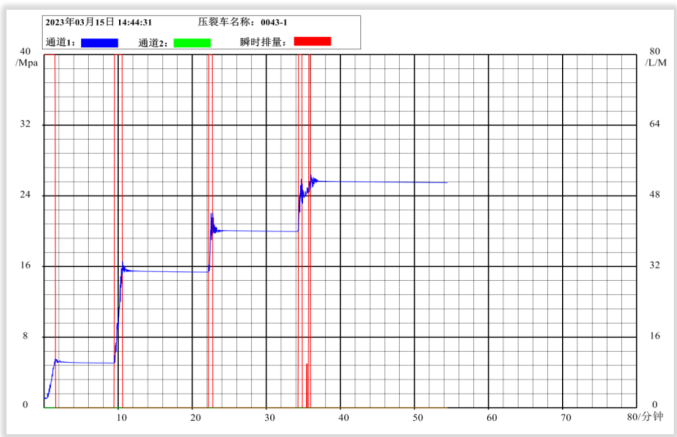


Figure 12—External Pressure Test Curve

During the patching construction, the tool string was run smoothly. The 3.5in casing propellant-powered patching system operated normally under downhole conditions at a depth of 1790m: the ignition system, pull force output, and release mechanism worked properly, and the patching propellant burned normally downhole. After patching, a full wellbore pressure test was conducted at 25MPa, with pressure stabilized for 20min and a pressure drop of 0.1MPa, meeting the client's requirements.

The second test well was Well XXX-07. In July 2023, multi-parameter combination logging and MIT +MTT analysis revealed a pinhole at 2694.73m (see Fig. 13). To improve the wellbore integrity of this well and eliminate subsequent operation risks, the cable-transmitted propellant-powered patching technology was used to patch and reinforce the casing pinhole leakage area of Well XXX-07. After patching, a sealed pressure test was conducted on the wellbore, which met the wellbore integrity requirements.

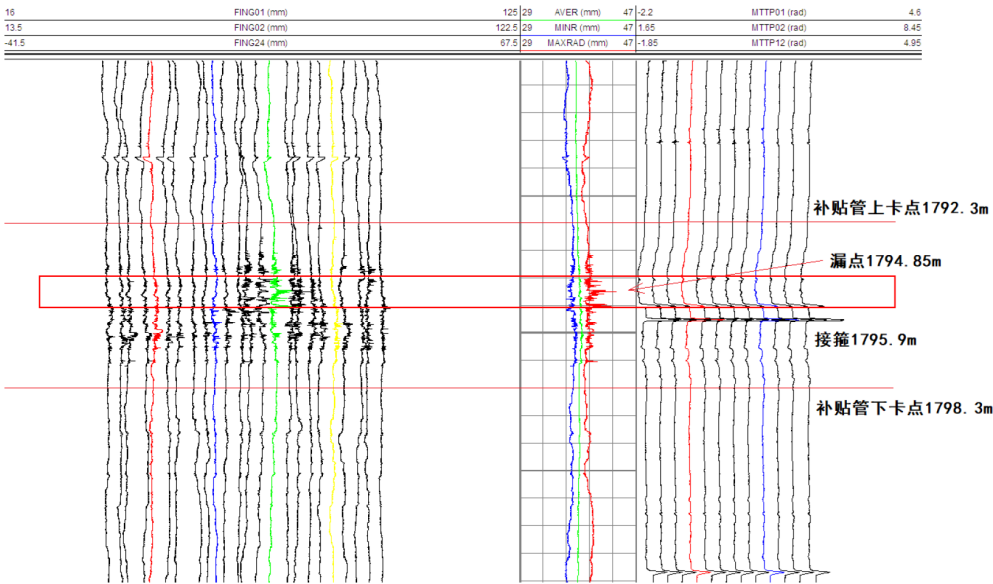


Figure 13—Schematic Diagram of Upper and Lower Sticking Points of Patching Tube

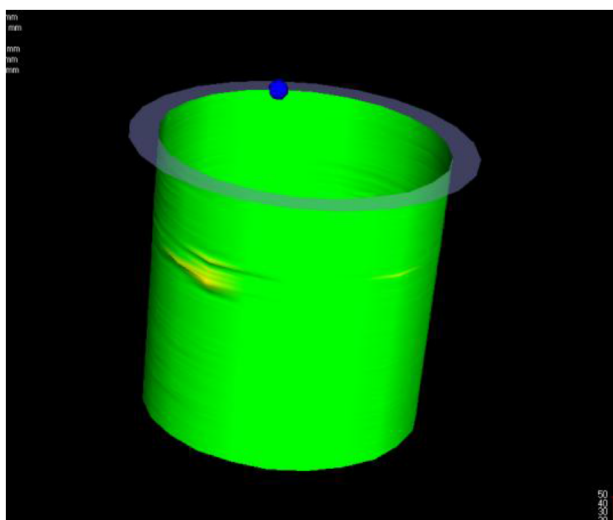


Figure 14—Imaging Diagram of Leak Point Location (2694.73m) in Well XXX-07

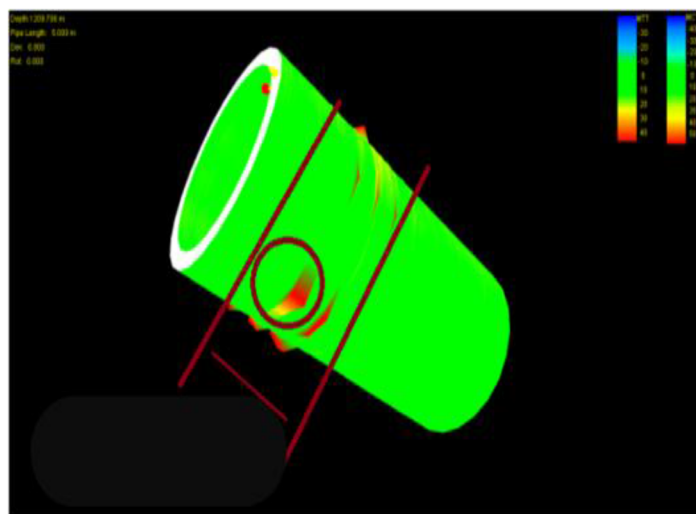


Figure 15—MIT thickness analysis

During the patching construction, the tool string encountered sticking twice while running into the well, but passed smoothly after pulling up and running down. The 3.5in casing propellant-powered patching system operated normally under downhole conditions at a depth of 2694.73m: the ignition system, pull force output, and release mechanism worked properly, and the patching propellant burned normally downhole. After patching, a full wellbore pressure test was conducted at 50MPa, with pressure stabilized for 10min and a pressure drop of 0.7MPa

Patch Parameters and Design

A 4-m patch tube was set to cover the leak. Top and bottom anchors were placed at 1 209.48 m and 1 213.48 m, respectively, 1 m above and 2 m below the damaged zone. The top anchor was 0.5 m below the coupling at 1 208.98 m. Parameters are given.

Operational Issues

Problem Analysis

During dummy runs with the patch tool + $\Phi 70\text{ mm} \times 4\text{ m}$ gauge, the tool tagged at 738 m. Overpull from 290 kg to 390 kg freed the tool, which was undamaged. Subsequent runs with $\Phi 70\text{ mm} \times 3\text{ m}$ gauge tagged at 845 m at 2 200–2 300 m/h. Finally, a $\Phi 68\text{ mm} \times 4\text{ m}$ gauge reached 1 250 m, but seven seal rings were scraped off.

Optimizations

Reduce friction: run three casing scrapers on coiled tubing from 800–1 300 m, followed by two clean-out runs with a flat-bottom mill shoe.

Resize tool: adopt $\Phi 68\text{ mm}$ gauge.

Reduce speed: lower POOH speed from 4 000 m/h to 1 500 m/h; no seal damage occurred.

Subsequent Operations

Surface Tests

$\Phi 68\text{ mm} \times 4\text{ m}$ and 3 m patch tubes were tested: external 40 MPa and internal 60 MPa held for 10 min without pressure drop.

Cable-Run Patch Operation

Install surface cable equipment. Assemble BHA: cable head + CCL + power tool + patch tube + guide. Run $\Phi 68\text{ mm} \times 3\text{ m}$ patch tube, slight tags at 887 m, reached 1 230 m successfully. Ignition data are shown in Tables 4–2 and 4–3. Cable tension dropped to 3.3 kN, tools retrieved intact; patch set.

Table 3—Simulation Calculation of Internal and External Pressure-Bearing Strength of Patched Tube After Patching

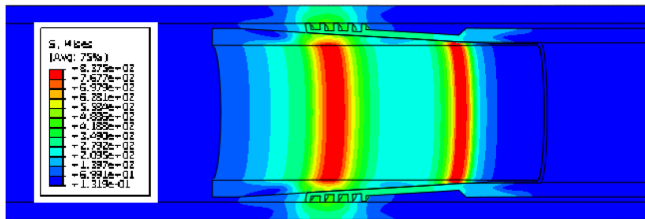
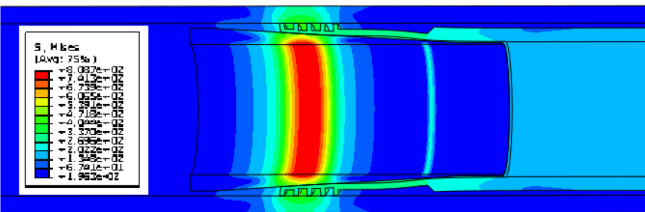
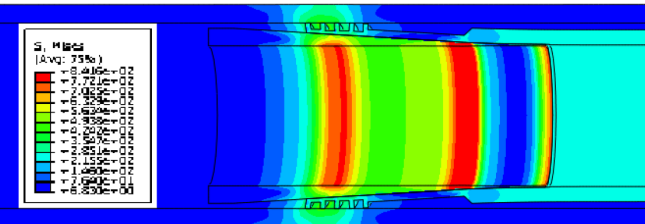
Name	Load	Simulation Result
Patching Tube Anchoring	Expansion Force: 250kN	
Patching Tube Internal Pressure Resistance	58.5MPa	
Patching Tube External Pressure Resistance	50MPa	

Table 4—Component Lengths

Component	length
CCL	0.452 m (cum. 0.452 m)
Power tool	3.17 m (3.622 m)
Top cone	0.115 m (3.737 m)
Patch tube	3.0 m (6.720 m)
Bottom cone	0.115 m (6.818 m)
Guide	0.12 m (6.938 m)

Table 5—Ignition Data

parameter	value
Bridge wire resistance	40 Ω
Cable resistance	56 Ω
Surface voltage	120 V
Ignition current	1.2 A
Result	normal ignition

Pressure Test

Stepwise pressure test to 55 MPa, 10 min pressure drop 0.3 MPa – successful. After parameter adjustments, the patch was qualified, subsequent fracturing was completed, and the well was put on production with a 60 000 m³/d gas rate.



Figure 16—Pressure Test Results After Casing Patching

Conclusion

Through the research and application of the 3.5in casing cable-transmitted propellant-powered patching technology, the following conclusions are drawn:

1. Through theoretical calculation, simulation, ground tests, and field applications, a 3.5in casing patching system suitable for cable transmission technology was designed.
2. The ground test of the propellant-powered patching tool was safe, and the tool operated normally. Third-party inspection of the patching simulation specimens showed that they could withstand an internal pressure of 61MPa (no leakage within 15 minutes) and an external pressure of 38MPa (no leakage within 15 minutes).
3. The 3.5in casing cable-transmitted propellant-powered patching technology was successfully applied for the first time in an oilfield. Through field applications, a set of construction plans for cable-transmitted propellant-powered casing patching was developed.

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